Fermionic Extensions of Continuous Functional Tensor Networks: Exchange-Rank and Exterior-Contraction Obstructions with Restricted Numerical Consequences

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Abstract

Hong et al. proposed continuous functional tensor networks by expanding each particle coordinate in an orthonormal functional basis, representing the coefficient tensor by a tensor network, evaluating basis-projected operators by contraction, and optimizing the tensors by automatic differentiation. We study the fermionic extension that their work left open. Two independent obstructions emerge. First, an ordinary particle-site tensor train pays an unavoidable binomial bond for exchange statistics; we recall the known flat particle spectrum of a single Slater determinant. Second, moving antisymmetry into exterior algebra removes that representation floor but does not make compact inputs easy to contract: exact squared-norm evaluation is #P-hard already at fixed one-form bond three under a polynomial-time Turing reduction. Along a matrix-pair subfamily, bounded and selected fixed-state memories collapse to polynomial linear combinations of established antisymmetrized geminal powers, while a bandwidth-one unique path already encodes a permanent. A generic relative squared-norm scheme would also approximate real positive-semidefinite permanents, whereas entrywise-nonnegative and additive promised cases remain open. Finally, cut-rank divisibility rejects the proposed universal direct tensor product of a statistics carrier and a Slater-normalized correlation multiplicity. Exact certificates and controlled ordered-coordinate calculations support the proof audit. A nonbranching exterior subclass is also evaluated as a numerical consequence of these constraints. Because that subclass is algebraically a finite nonorthogonal Slater expansion, the data are not claimed as a method distinct from nonorthogonal configuration interaction (NOCI). These exact obstructions constrain, but do not exclude, genuinely distinct restricted or controlled approximate algorithms; they do not establish a scalable generic fermionic solver.

1 Scope and thesis

The direct motivation is the functional-tensor-network construction of Hong–Xiao–Hu–Ji–Ran [1]. For orthonormal local functions $\{\phi_s\}_{s=1}^{D}$, a first-quantized wavefunction is written

$$\Psi(x_1, \ldots, x_N) = \sum_{s_1,\ldots,s_N} C_{s_1\cdots s_N} \prod_{i=1}^{N} \phi_{s_i}(x_i). \tag{1}$$

Orthonormality turns the continuum norm into the Euclidean norm of C and turns differential and multiplicative operators into finite functional-basis matrices. Hong et al. compressed C by an MPS and optimized its Rayleigh quotient with automatic differentiation; they explicitly named fermionic tensor networks as a future electronic extension.

For identical fermions, however, $C \in \Lambda^N V_D$. An ordinary MPS over the labelled particle slots and an occupation-number MPS use different tensor product factorizations. Their bond dimensions are not related by the abstract Hilbert-space isomorphism. Hilbert-space MPS [2], Grassmann/graded networks [3], and the ordered first-quantized construction of Li–Waintal [4] are therefore comparators, not aliases for the representation studied here.

Throughout, a *theorem* is a formal statement proved in the manuscript. Exact-arithmetic checks are reported only as bounded reproducibility checks and never replace a proof. Floating-point calculations are explicitly described as numerical evidence, with basis, truncation, and optimization controls. Unproved algorithmic consequences are called conjectures. Exact anti-symmetry is structural in every exterior or ordered-chamber construction below; its reported residual is $\epsilon_{\mathrm{AS}} = 0$. The generic exact-contraction obstruction motivates restricted and controlled approximate algorithms rather than excluding them.

The thesis is deliberately negative and representation-specific:

> Ordinary particle TT stores exchange multiplicity, whereas compact exterior cores can hide hard coefficient evaluation. Removing the former cost does not remove the latter, and neither observation is a no-go theorem for every fermionic tensor network, ordered sector, determinant expansion, or promised approximation.

2 Exchange rank in the labelled-particle factorization

Let V be a finite-dimensional real or complex Hilbert space. Identify $\Lambda^N V$ isometrically with the alternating subspace of $V^{\otimes N}$ by

$$J_N(v_1 \wedge \cdots \wedge v_N) = \frac{1}{\sqrt{N!}} \sum_{\pi \in S_N} \operatorname{sgn}(\pi) v_{\pi(1)} \otimes \cdots \otimes v_{\pi(N)}.$$

Write $C_{(k)}$ for the $k \mid N-k$ particle matricization.

Theorem 2.1 (Structural result I: exact particle-TT ranks)**.** *For $1 \le k < N$, the minimal exact ordinary-TT bond is*

$$r_k^{\mathrm{TT}}(C) = \operatorname{rank} C_{(k)}.$$

Proof. Any TT cut factors $C_{(k)}$ through its bond space, and successive exact singular-value decompositions attain all unfolding ranks [5]. □

Theorem 2.2 (Structural result II: universal exchange floor)**.** *If $0 \neq C \in \Lambda^N V$, then*

$$\operatorname{rank} C_{(k)} \ge \binom{N}{k}.$$

The lower bound is attained by every nonzero decomposable N-form.

Proof. Choose a nonzero coefficient c_I on an N-element support I. Restrict rows to k-subsets $A \subset I$ and columns to $I \setminus A$. The resulting $\binom{N}{k}$ square minor is diagonal up to signs: its diagonal entries are $\pm c_I$, while an off-diagonal entry repeats an index and vanishes. A decomposable form supported on an N-plane has no larger contraction image. □

Corollary 2.3 (strict-antisymmetry truncation floor)**.** *If $\widetilde{C} \neq 0$ and $r_k^{\mathrm{TT}}(\widetilde{C}) < \binom{N}{k}$ at any cut, then $\widetilde{C} \notin \Lambda^N V$.*

The TT-rank characterization is standard. The exterior rank-floor proof is included for audit, but no priority claim is made pending a dedicated exterior- algebra literature review. It is distinct from the stronger, generally open entropy-minimization statements studied for fermionic reduced density matrices [6, 4].

Theorem 2.4 (Structural result III: known flat Slater particle spectrum)**.** *For orthonormal* $u_1, \ldots, u_N$*, the normalized Slater determinant* Φ *has Schmidt decomposition*

$$\Phi = \frac{1}{\sqrt{\binom{N}{k}}} \sum_{|I|=k} (-1)^{\epsilon(I)} \Phi_I \otimes \Phi_{I^c}.$$

Thus all $\binom{N}{k}$ *nonzero singular values equal* $\binom{N}{k}^{-1/2}$*. The best cut-rank-r approximation has relative squared error*

$$\epsilon_{k,r}^2 = 1 - \frac{r}{\binom{N}{k}},$$

so error at most ϵ *requires* $r_k^{\mathrm{TT}} \geq \lceil (1-\epsilon^2)\binom{N}{k} \rceil$.

Proof. Partition the signed permutation sum by the set of orbitals in the first k slots. Each class factors into normalized internal Slater sums with coefficient $\sqrt{k!(N-k)!/N!}$. Distinct subsets give orthonormal states on both sides. The approximation formula is the Eckart–Young theorem applied to this flat singular-value list. This is the standard particle-reduced-density-matrix structure of a Slater determinant [7]. □

No flat spectrum is asserted for a general correlated alternating tensor. Theorems 2.1, 2.2, and 2.4 concern labelled-particle TT only; they do not apply to mode MPS, ordered chambers, or exterior/determinant carriers.

For a fixed orthonormal one-particle basis of size D, all of these representations describe states in exactly the same finite physical space $\Lambda^N V_D$, whose full-configuration-interaction dimension is $\binom{D}{N}$. A mode-space FCI or quantum-chemistry DMRG calculation in that orbital basis changes the factorization and variational compression, not the physical finite-basis space [8, 2]. Consequently, FEMPS has neither a larger variational space nor an accuracy or efficiency advantage merely by being first quantized. Such an advantage would require matched basis, tolerance, timing, and memory comparisons.

3 Exterior matrix-product representation

Definition 3.1 (one-form matrix-wedge state)**.** For site j, let $A^{[j]} \in \mathrm{Mat}_{\chi_{j-1}\times\chi_j}(V_D)$ with $\chi_0 = \chi_N = 1$. For compatible matrices of forms define

$$(X \wedge Y)_{ac} = \sum_b X_{ab} \wedge Y_{bc}.$$

The open-boundary state is

$$C(A^{[1]}, \ldots, A^{[N]}) = [A^{[1]} \wedge \cdots \wedge A^{[N]}]_{11} \in \Lambda^N V_D. \tag{2}$$

Exterior associativity makes matrix-wedge multiplication associative, and each virtual path is a wedge of N one-forms. Hence the state is alternating by construction and $\epsilon_{\mathrm{AS}} = 0$. The $\chi = 1$ family is exactly the decomposable forms. A sum of R Slaters embeds by R diagonal virtual paths, and the ordinary scalar MPS gauge action remains valid. These elementary facts do not supply a canonical form or an efficient norm contraction. The displayed parameter count $O(ND\chi^2)$ is an input-size statement only.

4 Fixed-bond exact contraction obstruction

For an $n \times n$ matrix $M = (M_{ij})$ over a noncommutative algebra, define the row-ordered Cayley determinant

$$\mathrm{CDet}(M) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) M_{1,\sigma(1)} \cdots M_{n,\sigma(n)}.$$

Chien–Harsha–Sinclair–Srinivasan prove #P-hardness over $\mathrm{Mat}_2(\mathbb{F})$ in characteristic zero [9]; related hardness results for the noncommutative determinant are given by Arvind–Srinivasan [10].

Lemma 4.1 (direct Cayley coefficient). *Let $M_{ij} \in \mathrm{Mat}_d(\mathbb{F})$ and set $\mathcal{A}^{[i]} = \sum_j M_{ij} e_j$. For $u, v \in \mathbb{F}^d$,*

$$u^\mathsf{T} \mathcal{A}^{[1]} \wedge \cdots \wedge \mathcal{A}^{[n]} v = \big(u^\mathsf{T}\, \mathrm{CDet}(M) v\big) e_1 \wedge \cdots \wedge e_n.$$

After absorbing the boundaries into the endpoint cores, this is a FEMPS with $D = N = n$ and maximum internal bond d.

Proof. A nonzero top-form term chooses every physical basis vector once. The choice is a permutation, its exterior sign is the permutation sign, and its virtual matrix product remains in increasing site order. Their sum is $\mathrm{CDet}(M)$. □

Theorem 4.2 (Fixed-bond squared-norm hardness). *Over $\mathbb{Q}$, exact squared-norm evaluation for rational one-form FEMPS is #P-hard under polynomial-time Turing reductions even when $D = N$ and all internal bonds are at most three. A generic exact contraction algorithm for fixed $\chi \leq 3$ would imply* $\mathrm{FP} = \#\mathrm{P}$.

Proof. Take a Cayley-determinant instance over $\mathrm{Mat}_2(\mathbb{Q})$. Standard boundary vectors select an entry x in Eq. (4.1); the resulting bond-two top-form state has squared norm x^2. The bond-one state $\Theta = e_1 \wedge \cdots \wedge e_n$ has coefficient one. A block-diagonal virtual direct sum represents $\Phi_x + \Theta$ with maximum bond three and norm $(x+1)^2$. Two norm queries recover the sign and value through

$$x = \frac{(x+1)^2 - x^2 - 1}{2}.$$

Four boundary pairs recover the 2×2 matrix output. Core construction, query count, and rational bit lengths are polynomial in the source input. □

The sharp bond boundary must be stated carefully. The Cayley amplitude state in the proof already has maximum bond $\chi = 2$ and its norm is x^2. The proved signed-amplitude recovery, however, adjoins the scalar reference Θ and therefore uses maximum bond three. Consequently the theorem above proves exact squared-norm hardness for fixed $\chi \leq 3$; it does *not* prove the corresponding statement with the promise $\chi = 2$.

Conjecture 4.3 (Bond-two exact squared-norm boundary). *Exact squared-norm evaluation remains #P-hard for rational one-form* FEMPS *whose maximum internal bond is two.*

Conjecture 4.3 is a precise open sharpening, not a theorem or an exact certificate. The established $\chi = 2$ fact is the pointwise hard amplitude encoding; the established norm theorem has maximum bond three.

An independent integer verifier checks the site/physical permutation sum, all four boundary entries, the explicit bond-three direct sum, polarization, and $\epsilon_{\mathrm{AS}} = 0$ for $2 \leq n \leq 6$. Its SHA-256 is `1d2208d3e5cb14f5c8e6c875f7fddf51c47ce9a3e61be6cedb8246d662b3a016`.

The result is not in conflict with the polynomial collapse of a fixed Mat_2 homogeneous pair power below. The hard one-form cores are site-dependent and preserve row order. Repeating one even two-form instead symmetrizes factor order.

The direct coefficient identity is pointwise algebraic and therefore also holds when the entries of M are one-particle functions evaluated at a coordinate. We do not infer a functional-input complexity theorem from that observation alone. Such a theorem additionally needs a polynomial-bit model for representing, evaluating, and interpolating the chosen functional basis. Absent those hypotheses, that complexity transfer remains a conjectural extension rather than a corollary of the finite rational theorem.

4.1 Why parity is only part of the mechanism

The contrast above is not an odd-versus-even dichotomy. Site-dependent odd one-form cores retain the ordered virtual product and can encode the hard row-ordered determinant. Repetition of one homogeneous even two-form makes the physical factors commute and can collapse a fixed coefficient algebra. But scalar-bond odd one-form states are just Slater determinants and are easy, whereas the growing-width even sparse-path family below is hard. Parity explains the commutation step, while site dependence, homogeneity, virtual width, and algebraic closure decide the contraction problem. Splitting a general matrix-valued pair core into consecutive one-form cores can also increase the bond by a factor of order D; it is not a cost-free equivalence.

4.2 Independent tagged pair-power mechanism

For $N = 2M$, a matrix-valued two-form $\Omega_B = \sum_{p<q} B_{pq} e_p \wedge e_q$ defines

$$\Psi_M(B; l, r) = l^\mathsf{T} \frac{\Omega_B^M}{M!} r.$$

Shift tags $E_{i,i+1} \otimes M_{ij}$ annihilate all factor orders except increasing row order. This gives a separate polynomial-time Turing reduction with pair bond $2(n+1)$ and one-form bond $O(N^2)$. It is weaker than Theorem 4.2 for unrestricted one-form cores, but isolates growing noncommutative order memory inside the homogeneous pair-power subfamily. Its order-1–4 exact-certificate hash is `893077be401414cd810fa1154e618d37d83b58e077732801f2482b3716b2c0c0`.

5 The matrix-pair corridor

5.1 Bounded Wedderburn–radical memory

Theorem 5.1 (Bounded-algebra LC–AGP collapse)**.** *Let $\mathcal{A}$ be a finite-dimensional unital complex algebra with radical $R^d = 0$ and Wedderburn–Malcev decomposition*

$$\mathcal{A} = S \oplus R, \qquad S = \bigoplus_{a=1}^{q} \mathrm{Mat}_{p_a}(\mathbb{C}).$$

Put $p = \max p_a$ and $\rho = \dim R$. For any linear boundary functional λ, $\lambda(\Omega^M)/M!$ is an exact LC–AGP with

$$K \le \sum_{k=0}^{\min(d-1,M)} q^{k+1} \rho^k \binom{M + v_k - 1}{v_k - 1}, \qquad v_k = (k+1)p^2 + k.$$

For fixed p and d, this is polynomial in the explicit input dimensions.

Proof. Expand $\Omega = \Omega_S + \Omega_R$ into noncommutative words. A word with at least d radical factors vanishes. For a surviving number k, resolve the $k+1$ semisimple runs through the q blocks and each radical insertion in a basis of R, giving at most $q^{k+1}\rho^k$ structural choices. For one choice, even physical two-forms commute; after applying λ, the result is a homogeneous degree-M scalar polynomial in at most $v_k = (k+1)p^2 + k$ two-forms. In characteristic zero, Mth powers of linear forms span $\mathrm{Sym}^M(\mathbb{C}^{v_k})$. Each such power divided by $M!$ is one AGP, proving Eq. (5.1). The radical filtration, block expansion, and polarization count are written out in Appendix A. □

The 2×2 upper-triangular and full Mat_2 base cases have exact certificates through $M = 6$ and $M = 4$, respectively. The latter needs at most $\binom{M+3}{3}$ AGPs even though the row-ordered determinant over the same matrix algebra is hard. This theorem classifies a pair-power family; it does not give an algorithm for the site-dependent cores of Theorem 4.2.

5.2 Selected growing memories

Theorem 5.2 (Fixed-state graded LC–AGP collapse)**.** *Suppose the coefficient algebra embeds faithfully in*

$$\mathrm{Mat}_w\Big(\mathbb{C}[z_1,\ldots,z_g]/(z_1^{d_1},\ldots,z_g^{d_g})\Big),$$

with fixed w, g. Then every boundary contraction of $\Omega^M/M!$ is an exact LC–AGP with

$$K \le \prod_{j=1}^{g}[M(d_j-1)+1]\binom{M+w^2-1}{w^2-1}.$$

In particular, the one-generator algebra $\mathbb{C}[z]/z^d$ requires at most $M(d-1)+1$ AGPs.

Proof. Before quotienting, every entry of the Mth power has degree at most $M(d_j-1)$ in counter z_j. Tensor-product Vandermonde interpolation extracts all retained coefficients using the first product in Eq. (5.2). At one grid point, a boundary of a $w \times w$ matrix power is a homogeneous degree-M polynomial in w^2 commuting two-forms; polarization supplies the second factor. Rational nodes give an exact rational construction for rational inputs. □

The theorem includes a noncommutative alternating-word algebra of dimension $2d-1$ embedded in $\mathrm{Mat}_2(\mathbb{C}[z]/z^d)$. Thus growing nilpotency and two path branches remain insufficient when automaton width and counter count are fixed. Weighted-automata and Waring/Veronese ingredients are established prior art; the project-specific result is only their LC–AGP classification.

5.3 A bandwidth-one hard path

Let $P_j = e_{2j-1} \wedge e_{2j}$ be orthonormal pair forms. Given $A \in \mathbb{F}^{M\times M}$, define edge forms $F_i = \sum_j A_{ij}P_j$ on the unique endpoint path of an upper-bidiagonal $(M+1)\times(M+1)$ pair matrix.

Theorem 5.3 (Sparse APG permanent obstruction)**.** *The endpoint state and its squared norm are*

$$\Psi_A = \frac{F_1\wedge\cdots\wedge F_M}{M!} = \frac{\mathrm{perm}(A)}{M!}P_1\wedge\cdots\wedge P_M, \qquad \langle\Psi_A,\Psi_A\rangle = \frac{\mathrm{perm}(A)^2}{(M!)^2}.$$

Exact squared-norm evaluation is therefore #P-hard for a family with pair bandwidth one, one virtual path, $D = 2M$, and $O(M^2)$ binary data.

Proof. Repeated P_j factors vanish. Every surviving multilinear term selects a permutation, and even pair forms commute, so its coefficient is the permanent. For a zero–one matrix, multiply the exact norm by $(M!)^2$ and take the nonnegative integer square root; Valiant's permanent theorem applies [11]. □

The state is an established antisymmetrized product of geminals, and permanent- valued APG/APIG coefficients and APG-to-AGP decompositions have direct prior art [12, 13]. The hard output is projectively a single top-form Slater, so this is an input-to-coefficient obstruction, not a lower bound on physical LC–AGP rank.

6 Relative approximation and energy certification

Exact permanent hardness alone does not exclude approximation. In particular, entrywise-nonnegative permanents admit an FPRAS [14]. The same sparse-path identity also accepts a different promise.

Theorem 6.1 (Real-PSD relative-norm transfer)**.** *Unless* RP = NP*, there is no randomized polynomial-time relative approximation scheme for the squared norm of every rational real upper-bidiagonal matrix-pair input, polynomial jointly in dimensions, input bit length, inverse tolerance, and logarithmic inverse failure probability.*

Proof. For rational real positive-semidefinite A, the sparse-path construction gives

$$s_A = \frac{\operatorname{perm}(A)^2}{(M!)^2}, \qquad \operatorname{perm}(A) \geq 0.$$

A relative estimate $\widetilde{s}_A$ would make $M!\sqrt{\widetilde{s}_A}$ a relative permanent estimate after a polynomial tolerance conversion. Meiburg excludes a PRAS for real-PSD permanents unless RP = NP [15]. $\square$

Additive permanent estimators remain possible [16]. They do not uniformly imply relative norm control: cancellations can make the true coefficient exponentially small in input precision while the additive scale remains order one.

Lemma 6.2 (Rayleigh-quotient certificate)**.** *If* $n > 0$*,* $E = h/n$*, and estimates obey*

$$|n - \widetilde{n}| \leq \Delta_n, \qquad |h - \widetilde{h}| \leq \Delta_h, \qquad \widetilde{n} > \Delta_n,$$

then, with $\widetilde{E} = \widetilde{h}/\widetilde{n}$*,*

$$|E - \widetilde{E}| \leq \frac{\Delta_h + |\widetilde{E}|\Delta_n}{\widetilde{n} - \Delta_n}.$$

For random bounds this holds on the simultaneous success event.

Proof. Use $E - \widetilde{E} = [(h - \widetilde{h}) - \widetilde{E}(n - \widetilde{n})]/n$ and $n \geq \widetilde{n} - \Delta_n > 0$. $\square$

Thus estimator unbiasedness alone is not an energy certificate; the denominator interval must stay positive. Tensor-network Monte Carlo [17] and sign-problem complexity [18] provide context but are not used as the reduction.

7 Failure of a universal direct statistics carrier

For $0 \neq C \in \Lambda^N V$, define the cut contraction

$$c_{C,k} : \Lambda^k V^* \longrightarrow \Lambda^{N-k} V, \qquad c_{C,k}(\alpha) = \iota_\alpha C,$$

and $\mathcal{B}_k(C) = \operatorname{im} c_{C,k}$. The tested proposal was an exact universal factorization

$$\mathcal{B}_k(C) \cong \mathcal{S}^{\text{fermion}}_{N,k} \otimes \mathcal{M}_k(C), \qquad \dim \mathcal{M}_k(\text{Slater}) = 1. \tag{3}$$

Theorem 7.1 (Cut-rank divisibility obstruction)**.** *For every* $N \geq 3$*, Eq.* (3) *cannot hold for every nonzero* N*-form.*

Proof. At the $1 \mid N-1$ cut, a Slater determinant has rank N, forcing $\dim \mathcal{S}^{\text{fermion}}_{N,1} = N$. Hence every admitted rank would be divisible by N. In $V = \mathbb{F}^{N+2}$, let

$$C_N = e_1 \wedge (e_2 \wedge e_3 + e_4 \wedge e_5) \wedge e_6 \wedge \cdots \wedge e_{N+2}.$$

Its $N+2$ one-index contractions have distinct monomial supports and are linearly independent, so $\operatorname{rank} c_{C_N,1} = N+2$. But $N \nmid N+2$. $\square$

The counterexample is only two Slaters. Shared physical covectors lock $N-2$ channels that an independent multiplicity tensor would treat as free. The rank is invariant under orbital basis changes and direct embeddings, and full one-cut rank is stable on an open neighborhood.

Standard symmetry-adapted tensor networks remain valid [19, 20]. Particle permutation supplies the one-dimensional sign representation, not a $\binom{N}{k}$ degeneracy. Under $GL(V)$, $\Lambda^N V$ is irreducible and the exterior cut has Littlewood–Richardson multiplicity one while retaining the full orbital representation. Hamiltonian-specific charges and multiplets are not ruled out. State-adaptive Slater/secant descriptions are established and need not be canonical [21, 22, 23].

The sparse-path construction sharpens the algorithmic warning: whenever its output is nonzero it is projectively one Slater, yet its scale is a permanent. A state-intrinsic multiplicity-one label cannot by itself compile compact cores into a norm algorithm.

8 Established positive controls and prior-art boundary

Fixed-number AGP is equivalent to number-projected pairing structure; projected BCS, HFB overlaps, and Pfaffian formulas are established [24, 25, 26]. AGP norms and reduced density matrices have polynomial algorithms [27]. Deterministic linear combinations of optimized and nonorthogonal AGPs precede this project [28, 29]; Pfaffian variational Monte Carlo with symmetry projection is also established [30, 31].

Accordingly, finite LC–AGP is used here as a validated tractable control, not as a new ansatz. The two collapse theorems above classify when matrix-pair memory reduces to that control. Grassmann signs or parity bookkeeping enforce fermionic algebra but do not, without additional closure, refute either hardness construction.

Lemma 8.1 (Explicit AGP embedding in one-form FEMPS)**.** *Let a scalar two-form be in skew-normal form $\Omega_F = \sum_{a=1}^{r} w_a u_a \wedge v_a$. For $1 \le M \le r$,*

$$\frac{\Omega_F^M}{M!} = \sum_{a_1<\cdots<a_M} \left(\prod_{j=1}^{M} w_{a_j}\right) u_{a_1} \wedge v_{a_1} \wedge \cdots \wedge u_{a_M} \wedge v_{a_M}$$

has an exact one-form FEMPS *with every internal bond at most $r \le \lfloor D/2 \rfloor$.*

Proof. Use the channel label a as the virtual state. The first core emits $w_a u_a$, the next core emits v_a diagonally, and every later u core has only strictly upper-triangular transitions $a < b$ with entry $w_b u_b$; the following v core is diagonal. Contracting the final v core with the right boundary leaves exactly the strictly increasing channel sequences displayed above. Thus the construction has $\binom{r}{M}$ nonzero paths but bond r, and never enumerates the paths in its core representation. □

For odd particle number, a blocked orbital b gives $b \wedge \Omega_F^M/M!$ by prepending one bond-one core; no auxiliary subtraction is needed for this representation statement. This embedding is a prior-art control, not a new AGP ansatz. It also does not contradict the $O(\chi D)$ bond that can arise when a general matrix-valued pair core of bond χ is split into one-form sites: skew-normal scalar AGP structure is the special closure used by the lemma.

9 Numerical validation and algorithm-design consequences

The 2201 mechanism was first reproduced on four coupled harmonic oscillators with $(D, \chi) = (8, 16)$, complex128 arithmetic, and automatic differentiation on an NVIDIA RTX PRO 4000 Blackwell GPU. At coupling $\gamma = -0.5$, the variational energy 1.8787064643 lies 1.160×10^{-5} above the exact continuum value. Independent scans vary D, χ, and four random seeds; CPU/GPU energy and gradient parity pass. This reproduces the functional- basis/operator/MPS/AD mechanism, not every plotted point in the original paper and not a fermionic result.

After generic exterior contraction failed, an independent ordered-coordinate control was retained. In the chamber $x_1 < \cdots < x_N$, use the center of mass R and positive gaps $r_i = x_{i+1} - x_i$. The map has unit Jacobian; collision faces are Dirichlet, and a normalized antisymmetric state is recovered by sign extension with the $\sqrt{N!}$ chamber factor. Full-line and half-line orthonormal functional bases preserve the 2201 operator calculus, while the coordinate reorganization follows Li–Waintal. This route is therefore not called FEMPS and carries no priority claim.

At the controlled $N = 8, D = 12$ point, a staged local action keeps the measured local-solver peak below 1.1 GB. The production energy is 44.4528442233; its difference from a finite exterior $D = 14$ numerical reference is 7.174×10^{-3}. Fourier, quadrature, MPO-bond, optimizer, and basis controls are recorded separately. The $D = 14$ value is not a continuum bound, the $N = 2, 4, 6, 8$ sequence is not an asymptotic fit, and no electron–electron cusp claim follows from the softened Coulomb interaction. A future real- Coulomb study must compare basis dimension versus error against quantum- chemistry DMRG in the same orbital basis and a matched first-quantized comparator.

The structural results were subsequently used to choose, rather than reject, a restricted numerical control. A nonbranching diagonal-path subclass represents a sum of K Slater paths and evaluates determinant transition quantities in $O(K^2)$ path pairs without enumerating all virtual histories or materializing the alternating coefficient tensor. As numerical evidence, for four interacting one-dimensional harmonic fermions with a finite-rank separable interaction, blind optimizations at $(D, K) = (6, 4)$ agree with finite-basis FCI to 1.9×10^{-12} in energy over three seeds, while a truth-free $D = 6 \to 7$ continuation has maximum finite-basis error 8.1×10^{-5} and variance 1.1×10^{-3}. At $D = 6$, the error falls from 5.3×10^{-3} at $K = 1$ to 2.3×10^{-11} at $K = 2$; at $K = 4$, the continuum error decreases monotonically over $D = 5, 6, 7$. Norm errors are below 5×10^{-16} and structural and materialized antisymmetry residuals are zero. Two additional clean-source schedules select different determinant paths but give a combined final-energy spread of 2.035×10^{-9}, maximum same-basis CI error 2.523×10^{-9}, maximum variance 1.462×10^{-8}, and no optimization failures. These data are retained in this manuscript as a bounded numerical exercise. The subclass is exactly a finite nonorthogonal Slater sum, so the calculation does not establish an ansatz beyond NOCI, an explicit-correlation advantage, improved functional-basis D convergence, asymptotic scalability, or superiority over FCI, particle TT, Li–Waintal, or same-basis DMRG.

10 Logical boundaries of the results

Table 1 records transitions most likely to create an invalid claim. Every row is a manuscript constraint, not an empirical observation.

11 Reproducibility checks

Table 2 lists bounded checks. Each verifier imports neither the production tensor backend nor the statement it checks. Passing a bounded certificate is evidence against transcription errors; it is not a proof for arbitrary size.

The complete certificate suite is reproduced from the repository root by

```
python -m pytest tests/test_exact_certificates.py -q
```

The test module records every verifier–certificate pair and invokes each standalone verifier in audit mode; Table 2 records the independently enumerated range for every claim.

The corresponding full SHA-256 values are:

Transition	What is established	What is not inferred
Exact/approximate	The two hardness constructions use exact arithmetic; the relative-norm theorem separately treats a randomized scheme.	Exact #P-hardness alone does not rule out an FPRAS or additive estimator.
Affine/projective	Norm polarization recovers the signed affine scale; the sparse-path output is projectively one Slater.	A simple normalized state does not make its compact-input normalization easy.
State/input	All complexity bounds measure explicit rational cores and coefficient bit length.	Minimal intrinsic Slater rank is not an input decoder or contraction algorithm.
Representation	The exchange-rank results apply to labelled-particle TT; fixed-bond hardness applies to exterior one-form cores.	Neither theorem transfers automatically to mode MPS, ordered sectors, or every fermionic TN.
Homogeneous/site-dependent	The collapse results repeat one matrix-valued even form; the Cayley construction uses row-labelled one-form cores.	Fixed Mat_2 pair-power collapse does not make fixed-bond site-dependent Cayley cores easy.
Energy/norm	A simultaneous numerator/norm event with a positive denominator interval certifies energy.	Unbiased numerator and norm estimates do not make their ratio unbiased or certified.

Table 1: Mandatory proof and wording transitions.

- Fixed-bond norm:
`1d2208d3e5cb14f5c8e6c875f7fddf51c47ce9a3e61be6cedb8246d662b3a016`.
- Tagged pair power:
`893077be401414cd810fa1154e618d37d83b58e077732801f2482b3716b2c0c0`.
- Bounded algebra, T_2:
`f671c2c10376c39cfb8c223edafba370570b9c417e8b12bcaaeb2f0f66cf078c`.
Mat_2:
`74d2de4a2cedcbaf548cd4c9895d0ea4af48e6bb485b72966562e46277a6d20d`.
- Graded algebra, one counter:
`07a222de3f44ced1b3fe155638299fdb8443a54f3d9f36479b994f32f9f0fd55`.
Alternating words:
`082238ff6e6783b7533b3b2a59f3664beb1820794bcae573add98baf43370030`.
- Sparse path:
`dd72c1aaeb0bc2a6b9206992cde9099f2f568b7ff6c8ed8eb7e38d958f78e790`.
- Relative norm:
`c15e7ff268a962e2790004c7f63d47bedb53be0c887ab0241e671b7fe4ff3b16`.
- Carrier obstruction:
`06025f168a49c0ab857c2163103ffabcb56fb04cd1fed4df9120d25ef6bc60df`.

12 Limitations and next questions

The results do not establish that every exterior family is hard, that every tractable exterior family is LC–AGP, or that additive approximation with a certified polynomial norm lower bound is impossible. Promised positive, low-rank, strongly orthogonal, Pfaffian/Gaussian, integrable, or low-treewidth families remain admissible if their promise excludes the hard embeddings and their

Claim	Range	What the verifier checks
Fixed-bond norm	$2 \leq N \leq 6$	Direct Cayley coefficient, four boundaries, explicit bond-three sum, norm polarization, exact antisymmetry.
Tagged pair power	orders 1–4	Perfect matchings and every pair-factor order against the tagged row-ordered determinant.
Bounded algebra	T_2: $M \leq 6$; Mat_2: $M \leq 4$	Exact polynomial powers versus simplex interpolation and LC–AGP coefficients.
Graded algebra	$M, d \leq 4$; alternating words $M \leq 3, d \leq 4$	Every boundary coefficient, Vandermonde extraction, and nested matrix-power interpolation.
Sparse path	$1 \leq M \leq 6$	Unique path, exterior subset expansion, and permutation permanent route.
Relative norm	bounded rational cases	Positive, cancelling, precision-ill-conditioned, signed PSD, and Rayleigh interval controls.
Carrier obstruction	$3 \leq N \leq 8$	All cut ranks, channel locking, orbital permutation, embedding, and full-rank perturbation.

Table 2: Exact-certificate coverage.

state and energy errors are certified. A genuinely different categorical carrier is not excluded by the divisibility obstruction; it must state its exactness, reconstruction cost, and contraction theorem rather than assume a free direct product.

The structural results and all currently admitted restricted-solver numerics belong to this single manuscript. No separate FEMPS method paper is claimed from the diagonal-path data. A future method manuscript is opened only after a representation beyond finite NOCI demonstrates a matched, reproducible advantage—for example, explicit correlation that improves functional-basis D convergence, or a controlled comparison with Li–Waintal and same-basis DMRG that identifies a genuine accuracy, stability, or complexity tradeoff. Higher-dimensional alternating-form rank classification is not pursued unless a specific rank question determines such an algorithmic or physical decision.

OpenAI Codex (GPT-5 family, accessed in August–September 2026) was used under human direction for literature discovery, code and proof exploration, reproducibility checks, and language editing. The human author selected the research questions, inspected the cited sources, verified the symbolic and numerical outputs, revised every proof and claim, and accepts responsibility for the manuscript. The AI system is not an author.

13 Conclusion

The attempted fermionic completion of the 2201 functional tensor network has identified two costs that must not be conflated. Ordinary particle TT pays a binomial exchange bond even for a Slater determinant. Exteriorizing the state removes that representation charge, but generic exact squared-norm contraction is already hard at fixed one-form bond, while a direct universal carrier factorization fails dimensionally. The tested tractable matrix-pair memories reduce to established LC–AGP structure, and a unique sparse path already contains a permanent. The nonbranching K-path calculation is correspondingly a useful NOCI-equivalent numerical control, not yet an independent method contribution. The resulting map is an algorithm-design constraint, not a denial of numerical FEMPS: genuinely distinct restricted subclasses and error-controlled approximate methods remain open and must be judged by matched interacting benchmarks, D and correlation convergence, cost, uncertainty, and antisymmetry residuals rather than by parameter counts alone.

A Detailed bounded-algebra collapse proof

This appendix expands the algebraic steps used in the bounded-algebra collapse theorem. Let $E = \Lambda^{\mathrm{even}} V_D$. The wedge product makes E a commutative $\mathbb{C}$-algebra, although it has zero divisors. Write $\Omega = \Omega_S + \Omega_R \in \mathcal{A} \otimes E_2$, where $E_2 = \Lambda^2 V_D$.

Radical filtration. Because R is a two-sided ideal, a product containing k factors from R, with arbitrary elements of S between them, belongs to R^k. This follows inductively: $SR^j, R^j S \subseteq R^j$, and $R^i R^j \subseteq R^{i+j}$. Hence every word in the expansion of Ω^M with $k \geq d$ radical factors vanishes. For fixed $k < d$, every surviving word has a unique run-length description

$$\Omega_S^{m_0} \Omega_R \Omega_S^{m_1} \cdots \Omega_R \Omega_S^{m_k}, \qquad m_0 + \cdots + m_k = M - k,$$

where zero-length semisimple runs are allowed. Summing over all nonnegative run lengths is equivalent to retaining the homogeneous degree-M part of the commutative physical-form polynomial constructed below; no additional combinatorial factor is omitted because the virtual product order remains the displayed order.

Wedderburn blocks and radical basis. Let 1_a be the central identity of the block $S_a \cong \mathrm{Mat}_{p_a}(\mathbb{C})$ and choose a basis $r_1, \ldots, r_\rho$ of R. Insert $1_S = \sum_{a=1}^{q} 1_a$ independently into each of the $k+1$ semisimple runs and expand each radical factor as

$$\Omega_R = \sum_{b=1}^{\rho} r_b \, \eta_b, \qquad \eta_b \in E_2.$$

Thus the contribution with k radical occurrences is a sum over at most $q^{k+1}\rho^k$ choices $\tau = (a_0, b_1, a_1, \ldots, b_k, a_k)$. This is an upper bound: incompatible block transitions may vanish and repeated basis choices may combine.

Fix one τ. In matrix units of S_{a_j}, the corresponding projected semisimple form has at most $p_{a_j}^2 \leq p^2$ scalar two-form entries $x_{j,\mu\nu} \in E_2$. Together with the selected radical forms $\eta_{b_1}, \ldots, \eta_{b_k}$, the boundary value under λ is a homogeneous degree-M polynomial

$$P_\tau(x_{0,\mu\nu}, \ldots, x_{k,\mu\nu}, \eta_{b_1}, \ldots, \eta_{b_k}) \in E_{2M}$$

in at most $v_k = \sum_{j=0}^{k} p_{a_j}^2 + k \leq (k+1)p^2 + k$ commuting two-forms. Matrix multiplication and λ affect only the scalar coefficients of this polynomial. Nilpotence inside E can identify or kill monomials, but cannot increase the number of powers needed below.

Polarization into AGPs. Let $z_1, \ldots, z_v$ be formal commuting variables. Over characteristic zero, the Veronese powers $(t_1 z_1 + \cdots + t_v z_v)^M$ span $\mathrm{Sym}^M(\mathbb{C}^v)$. One direct proof differentiates $(t \cdot z)^M$ at $t = 0$: every monomial z^α, $|\alpha| = M$, is a linear combination of evaluations at finitely many rational points by multivariate finite differences. Equivalently, choose $\binom{M+v-1}{v-1}$ generic rational points; the resulting Veronese evaluation matrix is nonsingular because its determinant is a nonzero polynomial. Therefore any homogeneous degree-M polynomial in v variables is a linear combination of at most $\binom{M+v-1}{v-1}$ Mth powers.

Substitute the scalar two-forms for the z_i and apply the algebra homomorphism from the symmetric algebra to E. Every resulting term $L^M/M!$ is an AGP. For a fixed k, multiplying the power count by the at most $q^{k+1}\rho^k$ structural choices and summing over $0 \leq k \leq \min(d-1, M)$ gives Eq. (5.1). All steps use only the fixed Wedderburn decomposition, radical basis, matrix multiplication, and exact linear algebra. For fixed p and d, the displayed count is polynomial in M, q, and ρ, hence in the explicit algebraic input size.

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